COSC 6385 - Homework 1: Out-Of-Order (O3) Simulation

Nicholas Anderson

1 Problem Statement

For this assignment we were given sample python code which simulates out of order exeuctions using the tomasulo algorithm. This algorithm includes reorder buffer (ROB), register alias table (RAT), a load-store-queue, reservation stations, functional units, and pipelining. All of these features come together to facilitate out-of-order execution. We were tasked with modifying this code to allow for multiple instructions to be executed during one clock cycle. Additionally I implemented allowance for commiting multiple instructions per clock cycle, as that is something that we talked about during the lecture in which we went over a practice problem closely related to this homework.

Additionally the input and output described in the assignment pdf is different from what is currently implemented. Specifically the existing input cannot take in register values, number of reservation stations (RS), cycles in EX, cycles in MEM, number of functional units (FUs), ROB entries, and issue width from the input text file. The output does not output the final register and non-zero memory values. Therefore I added this functionality into the existing code.

2 Expanding Issue Width

For this assignment we expand issue width to be able to issue and commit up to 4 instructions in one cycle. Expanding the amount of commands which can be issued in a cycle can be done directly through main by modifying the issue stage (lines 90-105 in main.py). This is done by adding a for loop of range issue_width, and issuing an instruction each iteration as long as the issuing is valid (i.e. there is an open RS for the instruction) and there are still instructions to be issued.

Expanding issue with for commits as well is done through the `commit` function in commit.py. Again this is done simply by adding a for loop of range issue_width, and committing an instruction each iteration as long is the head pointer of the ROB is pointing at a instruction that is ready to commit. Since instructions must commit in order, as soon as the instruction which the head pointer points to (implemented as `ROB[0]`) is not ready to commit, we break from the fop loop so no further operations are committed and the current operation is not removed from the ROB.

3 Input Parser

An input parser is used to be read from the input file the structure of our o3-simulator, that being the number of RS' for each instruction type (integer adder, FP adder/multiplier, and load/stores), The number of cycles each FU takes to execute and store in memory (memory is only impacted by load/store instructions, all other instructions take 0 cycles in the MEM pipeline stage), number of ROB entries, the issue width, and initial memory and register values. This is implemented in input_parser.py though standard parser methods (i.e. reading the input line by line and using regular expressions to parse values). Parsed values are placed into the `SimulationInput`. A `UnitConfig` class is also used, which `SimulationInput` derives values from to save

the full information of the program. A `SimulationInput` instance is returned to main through `read_input(filename)` (function in input_parser.py) containing all the information required to run the simulation with the parameters given in the input file.

4 Test Cases/Results

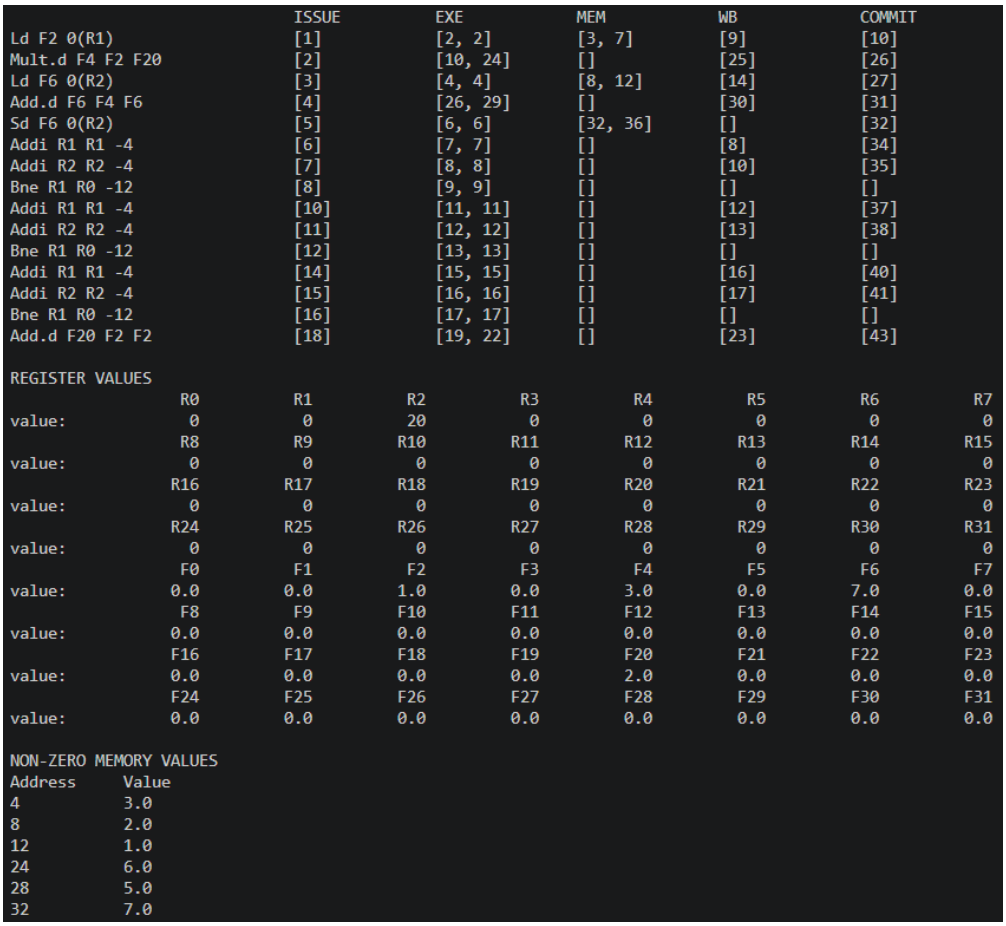
All test cases and results can be found in "test_cases" and "results" directories located in 'hw1/o3-simulator/'.

4.1 test_case.txt

The first test case we look at is the one given to us in test_case.txt, which has the following parameters for the input (issue width is not given as we change it to show how it impact runs):

Int. Adder: 4 RS, 1 Cycle in EX, 1 FU
FP Adder: 3 RS, 4 Cycles in EX, 1 FU
FP Mult.: 2 RS, 15 Cycles in EX, 1 FU
Load/Store: 5 RS, 1 Cycle in EX, 5 Cycles in Mem, 1 FU
ROB Entries: 64
Starting Register Values: R1=12, R2=32, F20=3.0
Starting Memory Values: Mem[4]=3.0, Mem[8]=2.0, Mem[12]=1.0, Mem[24]=6.0, Mem[28]=5.0, Mem[32]=4.0

The results of changing the issue width between 1,2,3, and 4 can be seen in figures 1, 2, 3, and 4. The main findings are that increasing issue width cannot increase the cycle of the final commit, however it can increase the cycle of intermediate commits. This is due to the way pipelining is implemented. Since EXE comes before MEM, instructions which do not require MEM will perform the WB step first since they are put into the results buffer during the `exe(...)` function call. Since the first instrution is a load operation, if non load/store instructions finish the EXE step on the same cycle that it finishes its MEM step, the non load/store instructions are placed into the results buffer first. This occurs when the issue width is 1, 3, and 4. Addi instructions finish the EXE stage on the same cycle which instruction 1 finishes its MEM stage. Since the Addi instructions (specifically their results) are placed into the results buffer first, they perform the WB stage first. Since only one instruction can perform the WB stage at one time, this means the first instruction has to wait for a later cycle. We can see by looking at 2 that an issue width of 2 avoids this, since instruction 6 finishes before cycle 7 (when instruction 1 finishes its MEM stage), and instructions 10 and 11 are not issued quickly enough to finish on cycle 7, the first instruction can perform the WB stage in the cycle directly after it finishes its MEM step.



	ISSUE	EXE	MEM	WB	COMMIT
Ld F2 0(R1)	[1]	[2, 2]	[3, 7]	[9]	[10]
Mult.d F4 F2 F20	[2]	[10, 24]	[]	[25]	[26]
Ld F6 0(R2)	[3]	[4, 4]	[8, 12]	[14]	[27]
Add.d F6 F4 F6	[4]	[26, 29]	[]	[30]	[31]
Sd F6 0(R2)	[5]	[6, 6]	[32, 36]	[]	[32]
Addi R1 R1 -4	[6]	[7, 7]	[]	[8]	[34]
Addi R2 R2 -4	[7]	[8, 8]	[]	[10]	[35]
Bne R1 R0 -12	[8]	[9, 9]	[]	[]	[]
Addi R1 R1 -4	[10]	[11, 11]	[]	[12]	[37]
Addi R2 R2 -4	[11]	[12, 12]	[]	[13]	[38]
Bne R1 R0 -12	[12]	[13, 13]	[]	[]	[]
Addi R1 R1 -4	[14]	[15, 15]	[]	[16]	[40]
Addi R2 R2 -4	[15]	[16, 16]	[]	[17]	[41]
Bne R1 R0 -12	[16]	[17, 17]	[]	[]	[]
Add.d F20 F2 F2	[18]	[19, 22]	[]	[23]	[43]

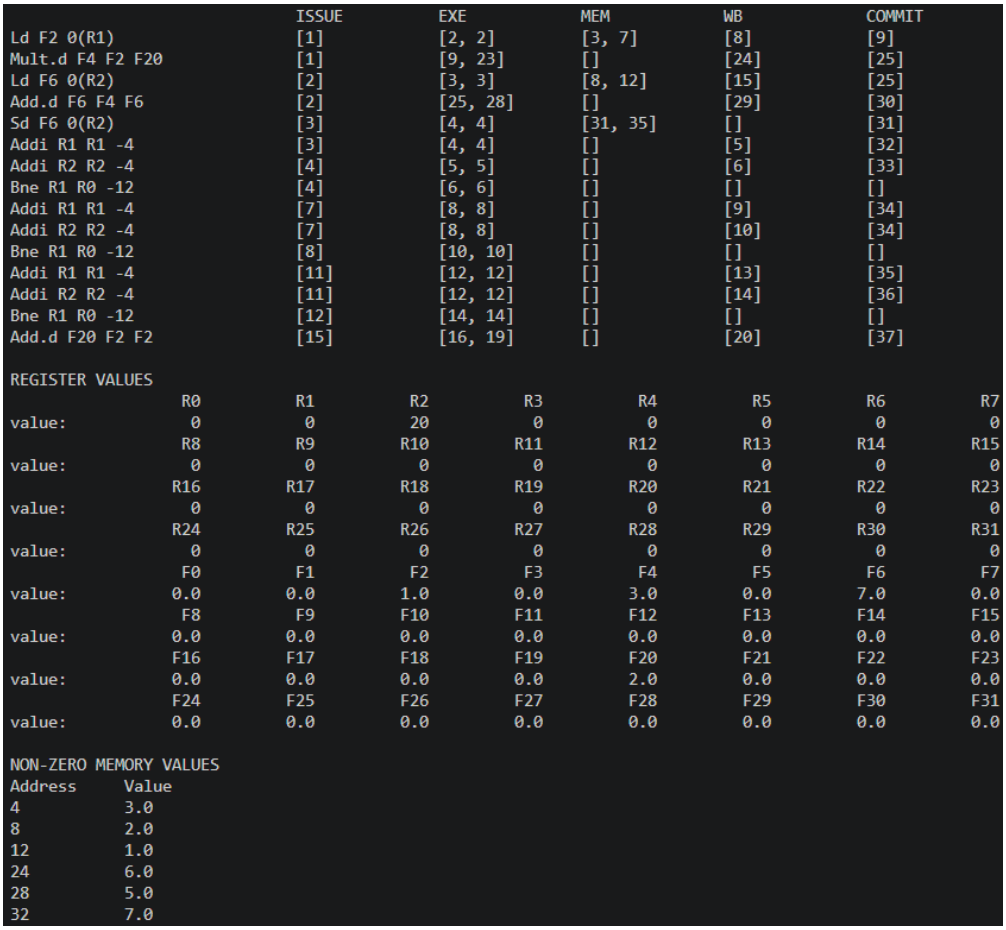
REGISTER VALUES

	R0	R1	R2	R3	R4	R5	R6	R7
value:	0	0	20	0	0	0	0	0
	R8	R9	R10	R11	R12	R13	R14	R15
value:	0	0	0	0	0	0	0	0
	R16	R17	R18	R19	R20	R21	R22	R23
value:	0	0	0	0	0	0	0	0
	R24	R25	R26	R27	R28	R29	R30	R31
value:	0	0	0	0	0	0	0	0
	F0	F1	F2	F3	F4	F5	F6	F7
value:	0.0	0.0	1.0	0.0	3.0	0.0	7.0	0.0
	F8	F9	F10	F11	F12	F13	F14	F15
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	F16	F17	F18	F19	F20	F21	F22	F23
value:	0.0	0.0	0.0	0.0	2.0	0.0	0.0	0.0
	F24	F25	F26	F27	F28	F29	F30	F31
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

NON-ZERO MEMORY VALUES

Address	Value
4	3.0
8	2.0
12	1.0
24	6.0
28	5.0
32	7.0

Figure 1: Results from test_case.txt with an issue width of 1.



	ISSUE	EXE	MEM	WB	COMMIT
Ld F2 0(R1)	[1]	[2, 2]	[3, 7]	[8]	[9]
Mult.d F4 F2 F20	[1]	[9, 23]	[]	[24]	[25]
Ld F6 0(R2)	[2]	[3, 3]	[8, 12]	[15]	[25]
Add.d F6 F4 F6	[2]	[25, 28]	[]	[29]	[30]
Sd F6 0(R2)	[3]	[4, 4]	[31, 35]	[]	[31]
Addi R1 R1 -4	[3]	[4, 4]	[]	[5]	[32]
Addi R2 R2 -4	[4]	[5, 5]	[]	[6]	[33]
Bne R1 R0 -12	[4]	[6, 6]	[]	[]	[]
Addi R1 R1 -4	[7]	[8, 8]	[]	[9]	[34]
Addi R2 R2 -4	[7]	[8, 8]	[]	[10]	[34]
Bne R1 R0 -12	[8]	[10, 10]	[]	[]	[]
Addi R1 R1 -4	[11]	[12, 12]	[]	[13]	[35]
Addi R2 R2 -4	[11]	[12, 12]	[]	[14]	[36]
Bne R1 R0 -12	[12]	[14, 14]	[]	[]	[]
Add.d F20 F2 F2	[15]	[16, 19]	[]	[20]	[37]

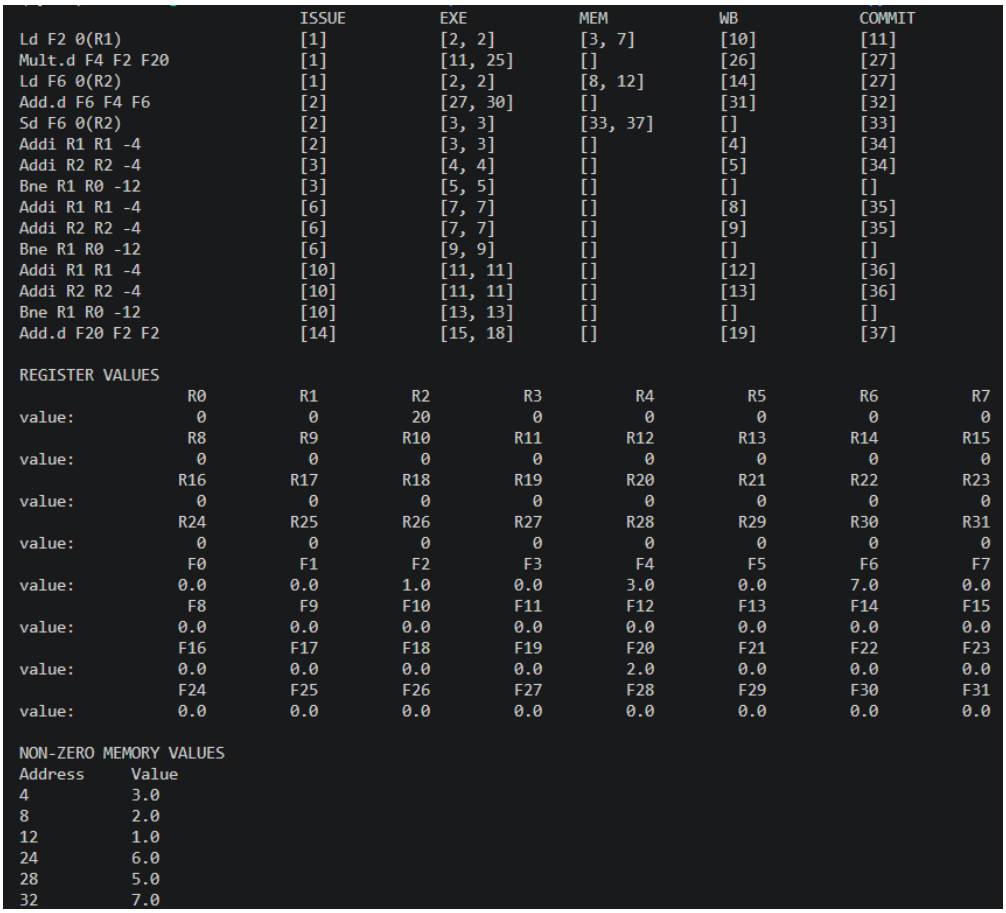
REGISTER VALUES

	R0	R1	R2	R3	R4	R5	R6	R7
value:	0	0	20	0	0	0	0	0
	R8	R9	R10	R11	R12	R13	R14	R15
value:	0	0	0	0	0	0	0	0
	R16	R17	R18	R19	R20	R21	R22	R23
value:	0	0	0	0	0	0	0	0
	R24	R25	R26	R27	R28	R29	R30	R31
value:	0	0	0	0	0	0	0	0
	F0	F1	F2	F3	F4	F5	F6	F7
value:	0.0	0.0	1.0	0.0	3.0	0.0	7.0	0.0
	F8	F9	F10	F11	F12	F13	F14	F15
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	F16	F17	F18	F19	F20	F21	F22	F23
value:	0.0	0.0	0.0	0.0	2.0	0.0	0.0	0.0
	F24	F25	F26	F27	F28	F29	F30	F31
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

NON-ZERO MEMORY VALUES

Address	Value
4	3.0
8	2.0
12	1.0
24	6.0
28	5.0
32	7.0

Figure 2: Results from test_case.txt with an issue width of 2.



	ISSUE	EXE	MEM	WB	COMMIT
Ld F2 0(R1)	[1]	[2, 2]	[3, 7]	[10]	[11]
Mult.d F4 F2 F20	[1]	[11, 25]	[]	[26]	[27]
Ld F6 0(R2)	[1]	[2, 2]	[8, 12]	[14]	[27]
Add.d F6 F4 F6	[2]	[27, 30]	[]	[31]	[32]
Sd F6 0(R2)	[2]	[3, 3]	[33, 37]	[]	[33]
Addi R1 R1 -4	[2]	[3, 3]	[]	[4]	[34]
Addi R2 R2 -4	[3]	[4, 4]	[]	[5]	[34]
Bne R1 R0 -12	[3]	[5, 5]	[]	[]	[]
Addi R1 R1 -4	[6]	[7, 7]	[]	[8]	[35]
Addi R2 R2 -4	[6]	[7, 7]	[]	[9]	[35]
Bne R1 R0 -12	[6]	[9, 9]	[]	[]	[]
Addi R1 R1 -4	[10]	[11, 11]	[]	[12]	[36]
Addi R2 R2 -4	[10]	[11, 11]	[]	[13]	[36]
Bne R1 R0 -12	[10]	[13, 13]	[]	[]	[]
Add.d F20 F2 F2	[14]	[15, 18]	[]	[19]	[37]

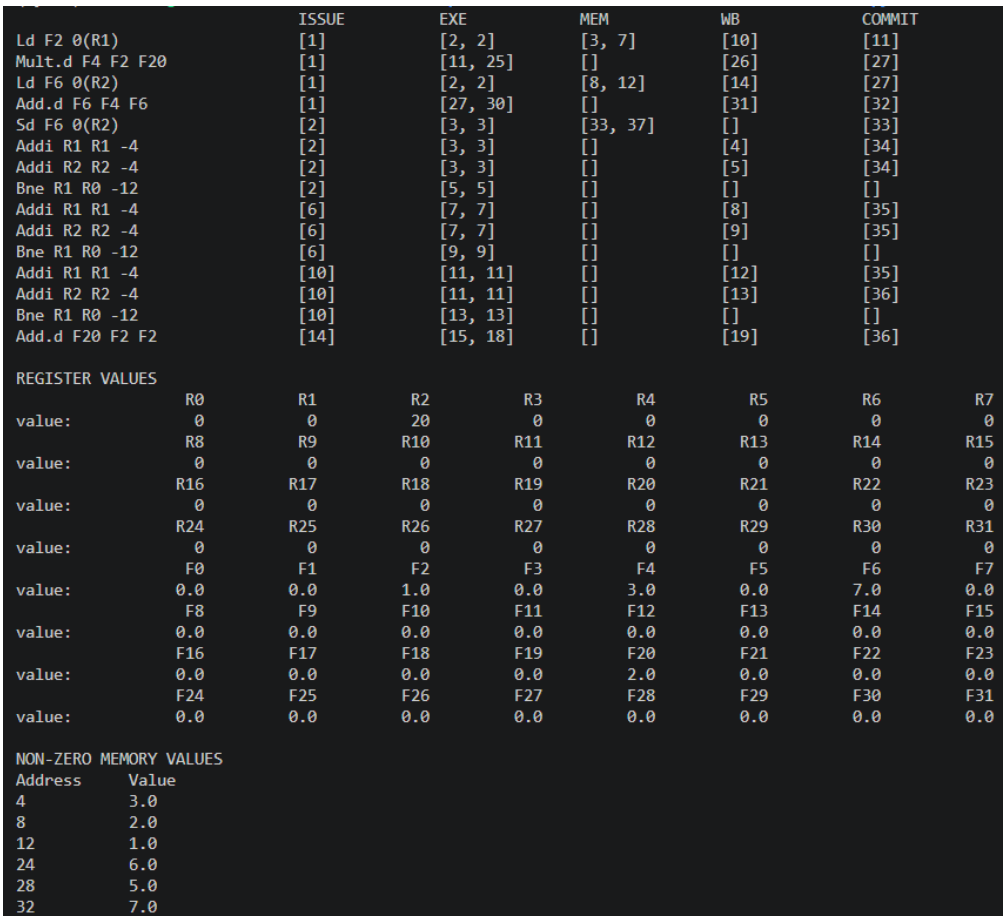
REGISTER VALUES

	R0	R1	R2	R3	R4	R5	R6	R7
value:	0	0	20	0	0	0	0	0
	R8	R9	R10	R11	R12	R13	R14	R15
value:	0	0	0	0	0	0	0	0
	R16	R17	R18	R19	R20	R21	R22	R23
value:	0	0	0	0	0	0	0	0
	R24	R25	R26	R27	R28	R29	R30	R31
value:	0	0	0	0	0	0	0	0
	F0	F1	F2	F3	F4	F5	F6	F7
value:	0.0	0.0	1.0	0.0	3.0	0.0	7.0	0.0
	F8	F9	F10	F11	F12	F13	F14	F15
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	F16	F17	F18	F19	F20	F21	F22	F23
value:	0.0	0.0	0.0	0.0	2.0	0.0	0.0	0.0
	F24	F25	F26	F27	F28	F29	F30	F31
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

NON-ZERO MEMORY VALUES

Address	Value
4	3.0
8	2.0
12	1.0
24	6.0
28	5.0
32	7.0

Figure 3: Results from test_case.txt with an issue width of 3.



	ISSUE	EXE	MEM	WB	COMMIT
Ld F2 0(R1)	[1]	[2, 2]	[3, 7]	[10]	[11]
Mult.d F4 F2 F20	[1]	[11, 25]	[]	[26]	[27]
Ld F6 0(R2)	[1]	[2, 2]	[8, 12]	[14]	[27]
Add.d F6 F4 F6	[1]	[27, 30]	[]	[31]	[32]
Sd F6 0(R2)	[2]	[3, 3]	[33, 37]	[]	[33]
Addi R1 R1 -4	[2]	[3, 3]	[]	[4]	[34]
Addi R2 R2 -4	[2]	[3, 3]	[]	[5]	[34]
Bne R1 R0 -12	[2]	[5, 5]	[]	[]	[]
Addi R1 R1 -4	[6]	[7, 7]	[]	[8]	[35]
Addi R2 R2 -4	[6]	[7, 7]	[]	[9]	[35]
Bne R1 R0 -12	[6]	[9, 9]	[]	[]	[]
Addi R1 R1 -4	[10]	[11, 11]	[]	[12]	[35]
Addi R2 R2 -4	[10]	[11, 11]	[]	[13]	[36]
Bne R1 R0 -12	[10]	[13, 13]	[]	[]	[]
Add.d F20 F2 F2	[14]	[15, 18]	[]	[19]	[36]

REGISTER VALUES

	R0	R1	R2	R3	R4	R5	R6	R7
value:	0	0	20	0	0	0	0	0
	R8	R9	R10	R11	R12	R13	R14	R15
value:	0	0	0	0	0	0	0	0
	R16	R17	R18	R19	R20	R21	R22	R23
value:	0	0	0	0	0	0	0	0
	R24	R25	R26	R27	R28	R29	R30	R31
value:	0	0	0	0	0	0	0	0
	F0	F1	F2	F3	F4	F5	F6	F7
value:	0.0	0.0	1.0	0.0	3.0	0.0	7.0	0.0
	F8	F9	F10	F11	F12	F13	F14	F15
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	F16	F17	F18	F19	F20	F21	F22	F23
value:	0.0	0.0	0.0	0.0	2.0	0.0	0.0	0.0
	F24	F25	F26	F27	F28	F29	F30	F31
value:	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

NON-ZERO MEMORY VALUES

Address	Value
4	3.0
8	2.0
12	1.0
24	6.0
28	5.0
32	7.0

Figure 4: Results from test_case.txt with an issue width of 4.